

Low Frequency Versus High Frequency PWM in Medium Voltage, High Power, Higher Level Inverters: THD, Harmonic Filtering, and Efficiency Comparison

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ABSTRACT One of the main advantages of multi-level inverters (MLI) is their ability to achieve high power quality and high efficiency power conversion. With the emergence of wide-band-gap (WBG) devices, the tendency has grown towards using high switching frequencies to improve converters' output power quality and minimize switching harmonic filters footprint. While high-frequency switching finds significant success in low and medium-power applications, it is not the best practice to reduce the filter size when applied to medium voltage (MV) high power (HP) inverters implemented with high voltage levels. In this paper, we show that the least filtering requirements for MV HP higher level inverters is achieved using the modulation strategy with the least switching, i.e., the staircase modulation. The paper compares the harmonic filter requirements and power losses for low-frequency and high-frequency pulse-width-modulated MLI. The analysis shows that low-frequency switching not only achieves the lowest losses, but also produces the lowest line-to-line voltage total harmonic distortion (THD), which allows eliminating both voltage and current harmonic filters in higher level inverters. Simulation results are provided to verify the theoretical analysis on the THD, and experiments on a 1 kW, 600Vdc MLI are presented to compare efficiency for low- and high-frequency modulation.

INDEX TERMS Switching frequency, total harmonic distortion (THD), medium voltage, multi-level inverters, harmonic filters, magnetics, pulse width modulation (PWM), efficiency.

I. INTRODUCTION

POWER electronics applications that demand ratings exceeding 250 kW require medium voltage (MV) converters to handle the high power (HP) levels. Nowadays, MV HP converters are widely used in heavy-duty electric and hybrid-electric propulsion/traction drives, as well as in other industrial applications including refiners, fans, pumps, and rolling mills [1], [2]. In addition, the increased grid integration of renewable energy resources to achieve a more sustainable and cleaner energy has led to the proliferation of MV HP inverters in utility applications [3], [4]. MV inverters have played a major role in modernizing traditional transmission and distribution

networks and facilitating the smart-grid transition through flexible ac transmission systems (FACTS) and resilient ac distribution systems (RACDS) [5]. MV converters can provide active control of power quality by minimizing voltage and current harmonics, provide volt/var/frequency compensation through shunt and series compensators [6], [7], [8], [9], [10], [11], [12], and achieve self-healing, islanding operation, fault handling and other functionalities [13], [14], [15], [16], [17], [18].

To achieve low common mode voltages (CMV), low dv/dt stresses, low electromagnetic interference (EMI), and better harmonic performance, MV HP inverters are implemented

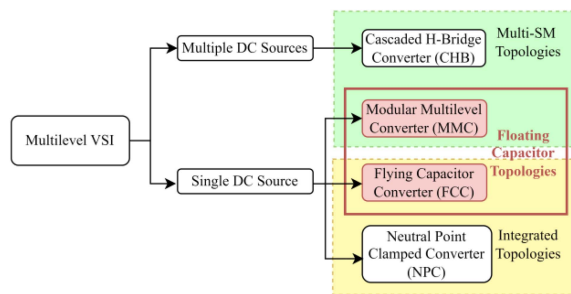


FIGURE 1. Multilevel VSIs classification.

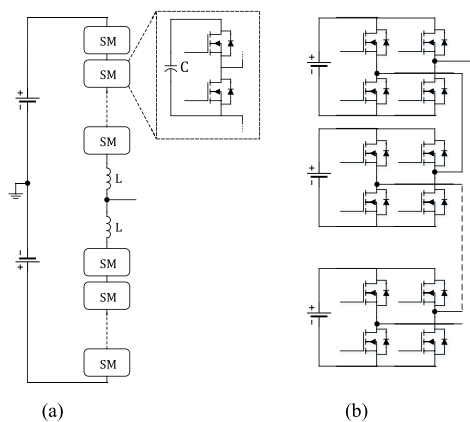


FIGURE 2. Higher level MLI topologies (a) MMC (b) CHB.

with multilevel topologies [19], [20], [21], [22], [23]. A general classification of multilevel voltage-source-inverter (VSI) topologies is presented in Fig. 1. Among those multilevel inverter (MLI) topologies, the modular multi-level converter (MMC) [24] shown in Fig. 2(a) and the cascaded H-bridge (CHB) inverter [6] shown in Fig. 2(b) are classified as higher level inverters, considering they are implemented with tens and even hundreds of levels.

Notwithstanding the benefits, utilizing MV converters presents some challenges, whether in grid or drives applications. These challenges are mainly driven by the need for more compact, more reliable, cheaper, faster, smaller, lighter and modular structures. Towards this goal, the first step was the evolution of the CHB inverters to achieve modularity compared to neutral point clamped (NPC) inverters and eliminate the bulky and lossy zig-zag transformers used with multi-pulse converters [1], [6], [10], [22]. This transition to CHB inverters added the modularity feature, and improved the dynamic response, efficiency, and cost. However, the need for isolated dc sources and the inability to have a back-to-back (B2B) connection with CHB inverters have led to the introduction of the MMC in 2002 [24]. The MMC has distinguished itself in MV HP applications due to its modularity and its ability to be used in B2B ac-dc-ac conversion systems [26], [27], [28], [29], [30], [31], [32], [33].



FIGURE 3. Three-phase passive harmonic filter bank.

Despite these topological improvements, many power density challenges still exist for utilizing MV inverters, and the use of bulky harmonic filters is one of them. In grid applications, PWM modulated inverters require inductive and capacitive line filters to eliminate the switching frequency voltage and current harmonics respectively and comply with THD requirements outlined in IEEE 519 standard [34]. Similarly, harmonic filters, in addition to EMI and common mode (CM) filters, are used in motor drive applications to reduce switching frequency ripples and common mode voltages (CMV), thus extend machine bearing life and insulation [35], [36]. These filters are bulky and lossy and deteriorate the MV conversion system's power density and efficiency. An example of a three-phase passive harmonic filter bank used in a MV distribution network can be seen in Fig. 3. It is obvious that shrinking the size of those passive filters or eliminating the need to use them is vital to improve the power density of MV conversion systems.

One of the main factors dictating the harmonic filter size is choice of the switching frequency of the inverter modulation scheme. With the emergence of wide-band-gap (WBG) devices enabling switching frequencies up to hundreds of kilohertz, the trend has been to push to high pulse-width-modulation (PWM) frequencies to minimize the inverters' output voltage and current harmonic filters footprint. This approach is widely used in low and medium power applications [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49].

Nevertheless, some questions arise in the following context: does maximizing the PWM frequency minimize magnetics footprint in MV inverter applications also? which begs the question, is high frequency PWM the way to go in MV MLI for lowest harmonic distortion? It turns out that the choice of the PWM frequency for MV high power inverters with high number of levels is a debatable topic in the power electronics community. In this paper, we address this debate and answer the questions above. First, we compare harmonic filter requirements for MV MLI under low and high frequency multi-level modulation. Theoretical analysis is carried out to quantify the line-to-line voltage total harmonic distortion (THD) for different number of voltage levels for both modulation approaches. The analysis then concludes on which modulation technique results in the minimum filtering

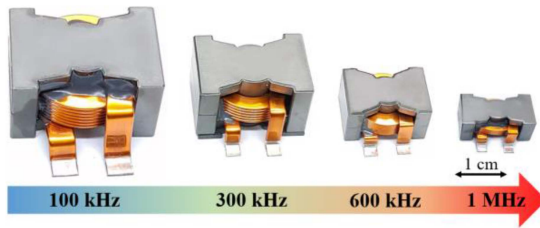


FIGURE 4. Inductor size reduction with switching frequency in SMPS.

requirements, specifically for MLI implemented with high number of levels, such as the CHB inverter or MMC. The study also analyzes the effect of modulation index on the THD. Simulations of a MMC with different number of voltage levels are presented to verify the THD analysis. Afterwards, the semiconductor switching and conduction losses are compared for low- and high-frequency modulation techniques, and experimental results on a 1 kW, 600Vdc, 4-level, single – phase MMC are presented to compare efficiency for the two modulation approaches.

The paper is organized as follows: In Section II, we present the dilemma between using the low frequency and high frequency PWM techniques. In Section III, the aforementioned theoretical analysis to compare the two modulation approaches is presented, backed up with simulation results. In Section IV, the power losses are compared for the two modulation approaches and experimental results are provided to compare efficiency. Finally, Section V concludes the paper.

II. THE HIGH- VERSUS LOW- MODULATION FREQUENCY DILEMMA

A. PWM FREQUENCY CHOICE FOR LOW AND MEDIUM POWER APPLICATIONS

There is no doubt that low and medium power general-purpose inverters and dc-dc converters have been improving in their performance by means of higher PWM frequencies. In inverters used in low and medium power motor drive applications, the inductive nature of the motor windings acts as a passive harmonic filter. As a result, lower inverter output current ripples are observed with higher PWM frequencies in these applications [46], [47]. Moreover, in switched mode power supply (SMPS) and automotive applications, higher switching frequencies have improved dc-dc converters' power density, owing to advancements in power semiconductor device technology. WBG power modules such as GaN and SiC devices enable switching frequencies up to few megahertz [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49]. With higher switching frequencies, the ripple-suppressing inductors will have a lower change in flux, which will allow a smaller range of flux density, and the use of a smaller inductor core. Moreover, differential-mode electromagnetic interference (EMI) filters can be designed with less inductance and capacitance, and smaller transformers can be used without saturating the magnetic core material [51]. Fig. 4

shows the change in the inductor core size with the increase in the switching frequency in low power applications.

Nevertheless, high frequency switching brings inevitable design challenges including high switching losses of the power semiconductors, and high magnetic losses due to skin and proximity effects, which lead to more thermal management challenges.

B. PWM FREQUENCY CHOICE FOR HIGH POWER HIGHER LEVEL INVERTER APPLICATIONS

As mentioned in the previous subsection, higher PWM frequency switching has enabled a substantial improvement in the power density of power electronic converters used in low and medium power applications. In fact, high switching frequency WBG devices have also revolutionized ac-ac conversion in MV applications, thanks to the solid state transformer (SST) technology [52], [53]. SSTs offer many benefits in smart grid and transportation applications, particularly the smaller magnetics form factor enabled by the medium and high frequency transformers.

However, some questions arise in the following context: does maximizing the PWM frequency minimize magnetics footprint in MV inverter applications also? which begs the question, is high frequency PWM the way to go in MV MLI for lowest harmonic distortion?

Looking at the literature [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], the link between the switching frequency and power density has been following one trajectory: pushing to high PWM frequencies to shift the harmonic spectrum of power electronic converters' output voltage and current to high frequencies, hence minimize the harmonic filter footprint. However, this approach, which finds a lot of success in low and medium power and SST applications, is recently expanding to MV high power inverters [54], [55], [56], [57], [58], [59]. This tendency arises nowadays due to the prevailing exploitation of WBG devices fast switching capabilities. Nevertheless, we will prove analytically in the next section that increasing the switching frequency increases voltage harmonic distortion and necessitates the use of passive filters, which would otherwise be eliminated with lowest frequency modulation, specifically when used for MV MLI implemented with high number of levels such as in the CHB inverter or MMC.

Unlike MV inverters, LV inverters are often implemented with few number of voltage levels; as a result, a PWM scheme with a high switching frequency is often used with capacitive and inductive filters at the output of the inverter. The LC filters provide passive filtering for the switching harmonics to achieve sinusoidal output voltage and current waveforms. However, MV inverters are usually implemented to synthesize multi-level output voltage waveforms. With tens of levels, higher level inverters, specifically MMC and CHB inverters, can generate a series of staircase voltage levels that approximates a sinusoidal voltage without having to do high frequency switching [1], [6], [10], [21], [60], [61]. As a result, the need for both voltage and current filters can be eliminated.

It is important to reiterate that low-frequency switching with high number of levels has been introduced for inverters long time ago [1], [6], [62]. Nevertheless, the core objective of this paper is to assess it from the harmonic filter sizing standpoint, especially with the rise of fast switching WBG devices nowadays, and their proliferation in MV high power applications.

III. LOW VERSUS HIGH PWM FREQUENCY – ANALYTICAL COMPARISON FROM THE TOTAL HARMONIC DISTORTION PERSPECTIVE

In what follows, we analytically compare the low frequency and the high frequency modulation approaches for MLI in terms of output voltage total harmonic distortion (THD). The analysis aims to give a picture on the harmonic filter requirements under different PWM frequencies, for MV inverters implemented with high number of levels as in the MMC and CHB inverters.

Two analyses are carried out, the first is for low frequency modulation (LFM) with multiple switching angles per quarter cycle and the other is for the high frequency pulse width modulation, represented by the two most common PWM strategies, which are the phase-shift PWM (PSPWM), and the level-shift PWM. The voltage THD is calculated theoretically for each of these modulation strategies for a given number of voltage levels, using the THD definition formula given in (1).

$$THD = \frac{1}{V_{a1}} \sqrt{V_{aRMS}^2 - V_{a1}^2}, \quad (1)$$

where V_{a1} is the fundamental component of the output phase voltage and V_{aRMS} is its rms value. In three-phase systems, the line-to-line voltage THD is more important than the phase voltage THD. For an $(N+1)$ -level multilevel converter, the number of voltage levels in the output phase voltage is $N+1$, and the number of voltage levels in the output line-to-line voltage is $2N+1$. In terms of harmonic content, the THD of a line-to-line waveform with M -levels (or steps) is the same as the THD of a phase waveform with M levels with the following exceptions:

- The line-to-line voltage waveform does not include the triplen harmonics (3^{rd} , 6^{th} , 9^{th} , 12^{th} ...).
- The line-to-line voltage waveform does not include the fundamental carrier frequency harmonics (only the sidebands).

Hence, the THD of the line-to-line voltage of an $(N+1)$ -level converter having $2N+1$ levels in its line-to-line waveform is derived by removing the triplen harmonics and the carrier frequency harmonics from the V_{a1} and V_{aRMS} equations where the variable N is set to $2N$.

For LFM, the switching angles are optimized to get the minimum THD for the line-to-line voltage, and the analysis is done for any odd number of switching angles per quarter fundamental cycle.

After the equations are developed, a plot showing the line-to-line voltage THD versus the number of output voltage levels for LFM with different number of switching angles

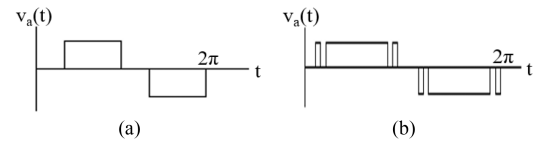


FIGURE 5. Phase voltage with (a) 1 switching angle per quarter cycle (LFM₁) (b) 3 switching angles per quarter cycle (LFM₃).

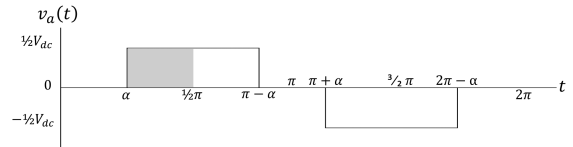


FIGURE 6. 3-level voltage waveform with 1 switching angle per quarter cycle (LFM₁). The shaded region represents 1 quarter cycle.

and for PSPWM and LSPWM is derived. For PSPWM and LSPWM, we further analyze the effect of carrier frequency choice on THD. The effect of modulation index is also studied. Finally, the modulation strategies are compared in terms of harmonic filter requirements.

A. LOW FREQUENCY MODULATION (LFM)

The first modulation strategy studied is the LFM. This strategy can also be divided into sub-strategies according to the number of switching angles per quarter cycle (or number of pulses) used. Using more switching angles per quarter cycle is essentially equivalent to increasing the switching frequency. In what follows, the LFM with p switching angles per quarter cycle, or LFM with p -pulses, is referred to as LFM_p. The LFM with 1-pulse (LFM₁) and 3-pulse (LFM₃) for a 3-level voltage waveform are shown in Fig. 5. LFM₁ is also known as fundamental frequency modulation or staircase modulation.

1) TRIGONOMETRIC FOURIER SERIES EXPRESSIONS

The first step to quantify the THD is to express the output phase a voltage of $(N+1)$ -level MLI in a trigonometric Fourier series (TFS) form. The 3 – level voltage waveform under LFM₁ (staircase modulation) is shown in Fig. 6. There is one switching angle α in the shaded region representing a quarter of a cycle. The TFS expression of this waveform is given in (2), where n is the harmonic number, V_{dc} is the total dc bus voltage, ω is the fundamental frequency in rad/s, and α is the switching angle.

$$v_a(t) = \frac{2V_{dc}}{\pi} \sum_{n=1}^{n=\infty} \frac{1}{n} \cos n\alpha \sin(n\omega t) \quad (2)$$

If the switching angle α is set to 0, the 3-level waveform reduces to the 2-level case, and the TFS for the 2-level case becomes as shown in (3)

$$v_a(t) = \frac{2V_{dc}}{\pi} \sum_{n=1}^{n=\infty} \frac{1}{n} \sin(n\omega t) \quad (3)$$

The TFS of the 3-level voltage waveform under LFM₁ can be extended to any odd-level or even-level voltage waveform. Let us define N as the number of voltage levels in the output phase voltage to neutral minus one. The TFS expression of the output phase voltage for an $(N + 1)$ -level MLI under LFM₁ is given in (4) shown at the bottom of this page, for the odd-level MLI case (N is even and $N > 1$), and in (5) shown at the bottom of this page, for the even-level MLI case (N is odd and $N > 1$).

The TFS expression obtained so far are for the LFM with 1 switching angle per quarter (LFM₁). This modulation technique is called the staircase modulation. The TFS expressions can be expanded to the LFM with multiple switching angles per quarter cycle. For better harmonic quality, the low frequency modulation techniques are often implemented with odd number of switching angles per quarter cycle only.

The TFS expression of the output phase voltage for the staircase modulated (LFM₁ modulated) 2-level MLI given in (3) can be generalized to any number of switching angles per quarter cycle p , where p is odd, as shown in (6) bottom of this page. Equation (6) shows the TFS of the output phase voltage for a 2-level MLI under LFM_p, where $\alpha_{x,y}$ represents the y^{th} switching angle of the x^{th} voltage level in rad.

Moreover, (6) can be generalized to the multilevel case. Equation (7) shown at the bottom of this page, shows the TFS expression for the odd-level MLI and (8) shown at the bottom of this page, shows the expression for the even-level MLI under LFM_p, where $N > 1$ and for any odd number of switching angles per quarter cycle p .

2) RMS EXPRESSIONS

The second step to quantify the THD is to find the rms of the output phase voltage. The rms of a periodic signal $f(t)$ is defined in (9), where T is the fundamental period of the signal,

over which the rms is computed.

$$\text{rms} = \sqrt{\frac{1}{T} \int_0^T f(t)^2 dt} \quad (9)$$

Using (9), the rms of the 3-level voltage waveform under LFM₁ (staircase modulation), is computed and given in (10) shown at the bottom of the next page.

The rms for the 2-level waveform under LFM₁ is obtained by setting α to 0 in (10), yielding $1/2 V_{dc}$. In fact, the rms of the 2-level waveform under LFM_p is also $1/2 V_{dc}$, regardless of the number of switching angles p used.

Next, the rms expression of the output phase voltage of any odd or even level MLI will be obtained. However, obtaining the rms expression by performing the integration using (9) is computationally complicated for MLI with voltage levels higher than 3. Hence, a geometrical approach is used whereby the rms of the output phase voltage is obtained by computing the area under the corresponding waveform. For instance, Fig. 7 shows the half-cycle waveform for the output phase voltage of an odd-level MLI under staircase modulation (LFM₁). The rms is two-fold the area under this waveform.

Equations (11) and (12) shown at the bottom of the next page, represent the rms expressions under LFM₁ for odd-level and the even-level MLI respectively ($N > 1$). Equations (13) and (14) are shown at the bottom of the next page, represent the rms expressions under LFM_p for odd-level and the even-level MLI respectively ($N > 1$), where p is odd and $p > 1$.

After obtaining the TFS and the rms expressions in terms of the switching angles, the latter are optimized to obtain the minimum line-to-line voltage THD according to the constraints presented in (15) and (16) shown at the bottom of the next page, where the THD is computed using (1). In (15) $q = 1/2N$ for the odd-level MLI and $q = 1/2(N - 1)$ for the

$$v_a(t) = \sum_{k=1}^{k=1/2N} \left\{ \frac{4V_{dc}}{N\pi} \sum_{n=1}^{n=\infty} \frac{1}{n} \cos n\alpha_k \sin n\omega t \right\} \quad (4)$$

$$v_a(t) = \sum_{k=1}^{k=1/2(N-1)} \left\{ \frac{4V_{dc}}{\pi N(N-1)} \sum_{n=1}^{n=\infty} \frac{1}{n} [\cos n\alpha_k (N-1) + 1] \sin n\omega t \right\} \quad (5)$$

$$v_a(t) = \frac{4V_{dc}}{\pi} \sum_{n=1}^{n=\infty} \sum_{m=1}^{m=p} \frac{(-1)^{m+1}}{n} \left(\cos n\alpha_{1,m} - \frac{1}{2} \right) \sin n\omega t \quad (6)$$

$$v_a(t) = \sum_{k=1}^{k=1/2N} \left\{ \frac{4V_{dc}}{N\pi} \sum_{n=1}^{n=\infty} \sum_{m=1}^{m=p} \frac{(-1)^{m+1}}{n} \cos n\alpha_{k,m} \sin (n\omega t) \right\} \quad (7)$$

$$v_a(t) = \sum_{k=1}^{k=1/2(N-1)} \left\{ \frac{4V_{dc}}{\pi N(N-1)} \sum_{n=1}^{n=\infty} \left(1 + \sum_{m=1}^{m=p} \frac{(-1)^{m+1}}{n} \cos n\alpha_{k,m} \right) \sin (n\omega t) \right\} \quad (8)$$

even-level MLI. In (16), V_{a1} is the desired fundamental voltage, which is equal to the reference voltage $V_a^* = 1/2 MI \times V_{dc}$, where MI is the modulation index.

B. HIGH FREQUENCY PSPWM AND LSPWM

The THD for the high-frequency modulated MLI shall be derived following the same approach as of the LFM case; i.e., finding the TFS and the rms expressions, followed by using (1).

1) DOUBLE TRIGONOMETRIC FOURIER SERIES EXPRESSIONS (DFTS)

The double trigonometric Fourier series expression (DFTS) of the output phase voltage of a 2-level inverter with double-edge naturally sampled PWM is given in [63]. Using this expression, the DFTS expression of the output voltage of the PSPWM modulated MLI is shown in (17) shown at the bottom of this page, and that for the LSPWM modulated MLI in (22) shown at the bottom of the next page. The A term represents the fundamental component and it is given by (18) shown at the bottom of this page. The B_m term represents the carrier harmonics and it is given by (19) shown at

$$V_{aRMS} = \frac{V_{dc}}{2} \sqrt{\frac{\pi - 2\alpha}{\pi}} \quad (10)$$

$$V_{aRMS} = \frac{2V_{dc}}{N} \sqrt{\frac{1}{2\pi} \left[\sum_{k=1}^{k=1/2N-1} \{k^2 (\alpha_{k+1} - \alpha_k)\} + \left(\frac{N}{2}\right)^2 \left(\frac{\pi}{2} - \alpha_{1/2N}\right) \right]} \quad (11)$$

$$V_{aRMS} = \frac{2V_{dc}}{N} \sqrt{\frac{1}{2\pi} \left[\sum_{k=1}^{k=1/2(N-1)-1} \left\{ \left(\frac{\frac{N-1}{2}-k}{N(N-1)} + \frac{k}{N-1} \right)^2 (\alpha_{k+1} - \alpha_k) \right\} + \left(\frac{\frac{N-1}{2}}{N-1} \right)^2 \left(\frac{\pi}{2} - \alpha_{1/2(N-1)} \right) + \left(\frac{\frac{N-1}{2}}{N(N-1)} \right)^2 \alpha_1 \right]} \quad (12)$$

$$V_{aRMS} = \frac{2V_{dc}}{N} \sqrt{\frac{1}{2\pi} \left[\sum_{k=1}^{k=1/2N-1} \left\{ (k+1)^2 \sum_{m=1}^{m=p-1} (-1)^m \alpha_{k+1,m} \right\} + k^2 \left(\alpha_{k+1,1} - \alpha_{k,p} + \sum_{m=2}^{m=p} (-1)^{m+1} \alpha_{k+1,m} \right) + \left(\frac{N}{2} \right)^2 \left(\frac{\pi}{2} - \alpha_{1/2N,p} \right) + \sum_{m=1}^{m=p-1} (-1)^m \alpha_{1,m} \right]} \quad (13)$$

$$V_{aRMS} = \frac{2V_{dc}}{N} \sqrt{\frac{1}{2\pi} \left[\sum_{k=1}^{k=1/2(N-1)-1} \left\{ \left(\frac{\frac{N-1}{2}-k}{N(N-1)} + \frac{k}{N-1} \right)^2 \sum_{m=1}^{m=p} (-1)^{m+1} \alpha_{k+1,m} + \left(\frac{\frac{N-1}{2}-k-1}{N(N-1)} + \frac{k+1}{N-1} \right)^2 \sum_{m=1}^{m=p-1} (-1)^m \alpha_{k+1,m} \right\} + \left(\frac{\frac{N-1}{2}}{N-1} \right)^2 \left(\frac{\pi}{2} - \alpha_{1/2(N-1),p} \right) + \left(\frac{\frac{N-1}{2}}{N(N-1)} \right)^2 \sum_{m=1}^{m=p} (-1)^{m+1} \alpha_{1,m} + \left(\frac{\frac{N-1}{2}-1}{N(N-1)} + \frac{1}{N-1} \right)^2 \sum_{m=1}^{m=p-1} (-1)^m \alpha_{1,m} \right]} \quad (14)$$

$$0 < \alpha_{1,1} < \alpha_{1,2} < \dots < \alpha_{1,p} < \alpha_{2,1} < \alpha_{2,2} < \dots < \alpha_{2,p} < \alpha_{q,1} < \alpha_{q,2} < \dots < \alpha_{q,p} < \frac{\pi}{2} \quad (15)$$

$$V_{a1} = V_a^* \quad (16)$$

$$v_a(t) = A \cos(\omega_0 t + \theta_0) + \frac{1}{N} \sum_{k=1}^{k=N} \sum_{m=1}^{m=\infty} B_m \cos \left[m \left(\omega_c t + \theta_c + \frac{2\pi(k-1)}{N} \right) \right] + \frac{1}{N} \sum_{k=1}^{k=N} \sum_{m=1}^{m=\infty} \sum_{\substack{n=-\infty \\ n \neq 0}}^{n=\infty} C_{mn} \cos \left[m \left(\omega_c t + \theta_c + \frac{2\pi(k-1)}{N} \right) + n(\omega_0 t + \theta_0) \right] \quad (17)$$

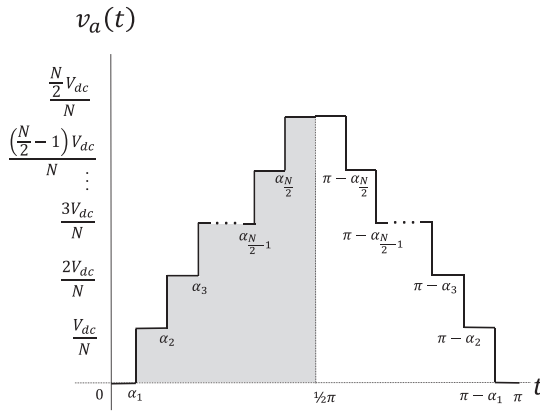


FIGURE 7. Half-cycle waveform for the phase voltage of an odd-level MLI under LFM₁. The shaded region represents one quarter cycle.

the bottom of this page. The C_{mn} term represents the sideband harmonics and it is given by (20) shown at the bottom of this . m and n are the harmonic numbers of the carrier and sideband harmonics respectively. ω_0 and ω_c represent the fundamental and carrier frequencies respectively in rad/s, and θ is an arbitrary phase shift angle of the fundamental waveform.

2) RMS EXPRESSIONS

The rms of the output phase voltage waveform is computed using (9), and it is given in (21) shown at the bottom of this page, for PSPWM and in (23) shown at the bottom of this page, for LSPWM.

C. THD VERSUS NUMBER OF LEVELS

In the previous sections, the TFS expressions and the rms equations of the output phase voltage of an $(N+1)$ -level MLI were given under high frequency PSPWM and LSPWM and under LFM with different number of switching angles per quarter cycle. Using (1), the line-to-line voltage THD can be computed and compared under these modulation methods, for different number of voltage levels. It is important to reiterate that using more switching angles per quarter cycle is essentially equivalent to increasing the switching frequency.

1) RESULTS ANALYSIS

Fig. 8 shows the line-to-line voltage THD versus the number of MLI voltage levels under low- and high-frequency modulation. For LFM modulation (LFM₁, LFM₃, LFM₅, LFM₇, and LFM_p), there are (1, 3, 5, 7, and p) switching instants per quarter cycle per voltage level. Hence, the switching frequency for LFM_p is $pf_{fund}MI$, where f_{fund} is 60 Hz, and MI is the modulation index. For phase shift PWM (PSPWM) and level shift PWM (LSPWM), the carrier frequency used is $f_c = 5$ kHz. For PSPWM, the effective switching frequency is the

$$A = \frac{MI \times V_{dc}}{2} \quad (18)$$

$$B_m = \frac{2V_{dc}}{\pi} \frac{1}{m} J_0 \left(\frac{m \times MI \times \pi}{2} \right) \sin \left(\frac{m \times \pi}{2} \right) \quad (19)$$

$$C_{mn} = \frac{2V_{dc}}{\pi} \frac{1}{m} J_n \left(\frac{m \times MI \times \pi}{2} \right) \sin \left[\frac{(m+n) \times \pi}{2} \right] \quad (20)$$

$$V_{aRMS} = \sqrt{\frac{1}{2} \left[\frac{MI \times V_{dc}^2}{4} + \frac{1}{N} \frac{V_{dc}^2 (2 - MI^2)}{4} + \frac{2}{N^2} \left[\sum_{m=1}^{\infty} \left(B_m^2 \sum_{l=1}^{N-1} \sum_{k=1}^{N-l} \cos [m (\theta_{N-k+1} - \theta_l)] \right) \right] + \frac{4}{N^2} \left[\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} C_{mn}^2 \left(\sum_{l=1}^{N-1} \sum_{k=1}^{N-l} \cos [m (\theta_{N-k+1} - \theta_l)] \right) \right] \right]} \quad (21)$$

$$v_a(t) = A \cos(\omega_0 t + \theta_0) + \frac{1}{N} \sum_{k=1}^{k=N} \sum_{m=1}^{m=\infty} \frac{B_m}{k} \cos [m (\omega_c t + \theta_c)] + \frac{1}{N} \sum_{k=1}^{k=N} \sum_{m=1}^{m=\infty} \sum_{\substack{n=-\infty \\ n \neq 0}}^{n=\infty} \frac{C_{mn}}{k} \cos [m (\omega_c t + \theta_c) + n (\omega_0 t + \theta_0)] \quad (22)$$

$$V_{aRMS} = \sqrt{\frac{1}{2} \left[\frac{MI \times V_{dc}^2}{4} + \frac{1}{N} \frac{V_{dc}^2 (2 - MI^2)}{4} + \frac{2}{N^2} \left[\sum_{m=1}^{\infty} \left(B_m^2 \sum_{l=1}^{N-1} \sum_{k=1}^{N-l} \frac{1}{k} \cos [m (\theta_N - \theta_l)] \right) \right] + \frac{4}{N^2} \left[\sum_{m=1}^{\infty} \sum_{n=1}^{\infty} C_{mn}^2 \left(\sum_{l=1}^{N-1} \sum_{k=1}^{N-l} \frac{1}{k} \cos [m (\theta_N - \theta_l)] \right) \right] \right]} \quad (23)$$

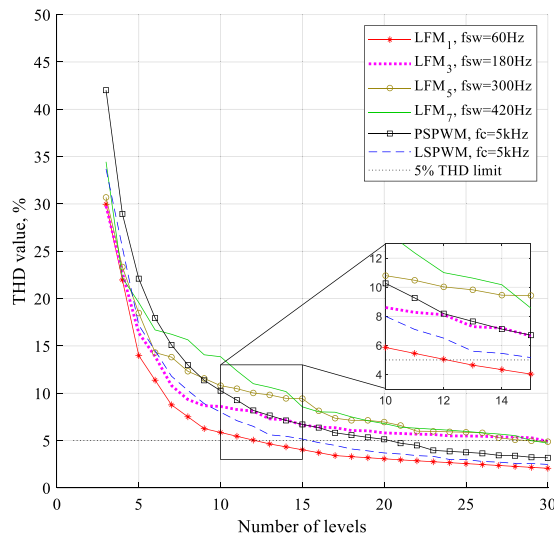


FIGURE 8. Line-to-line voltage THD versus the number of inverter voltage levels under low- and high-frequency modulation at unity modulation index.

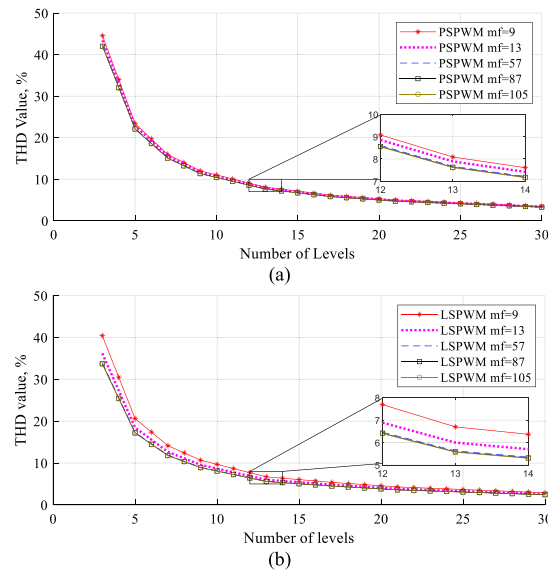


FIGURE 9. The influence of carrier frequency modulation index on line-to-line voltage THD for different number inverter voltage levels with (a) PSPWM (b) LSPWM.

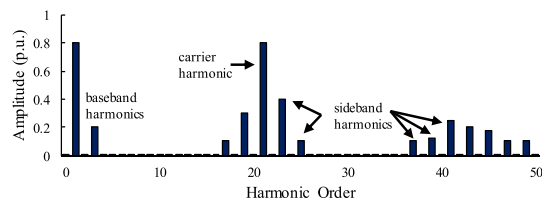


FIGURE 10. Output voltage harmonic spectra of carrier-based modulated inverter.

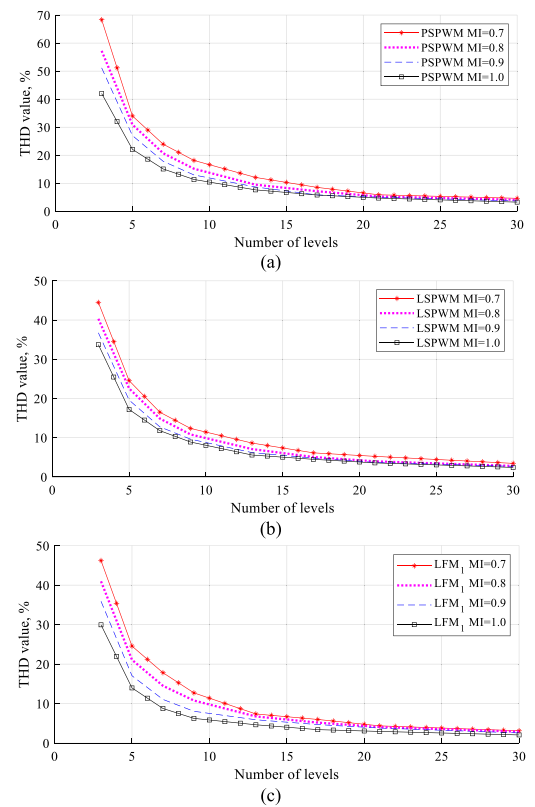


FIGURE 11. The influence of modulation index on line-to-line voltage THD for different number inverter voltage levels with (a) PSPWM (b) LSPWM (c) staircase modulation (LFM₁).

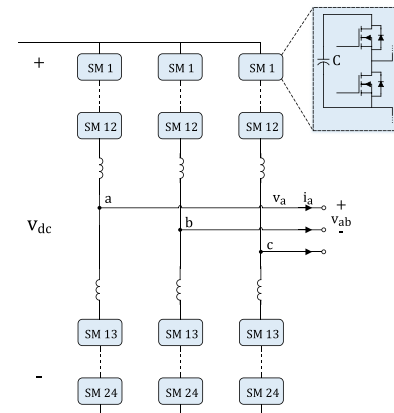


FIGURE 12. Simulated 13-level 3-phase MMC topology.

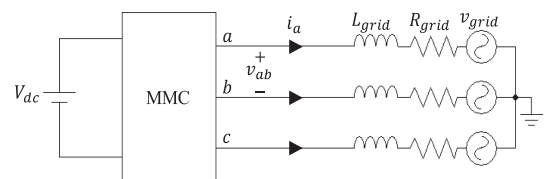


FIGURE 13. Circuit Diagram of the filter-less grid-connected MMC.

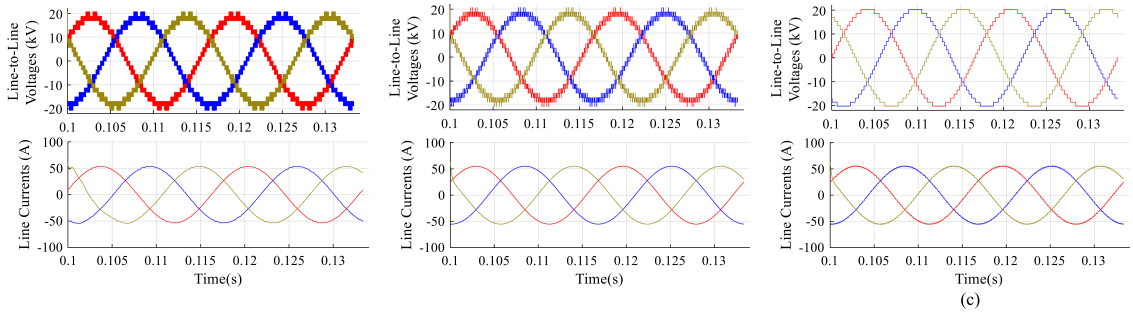
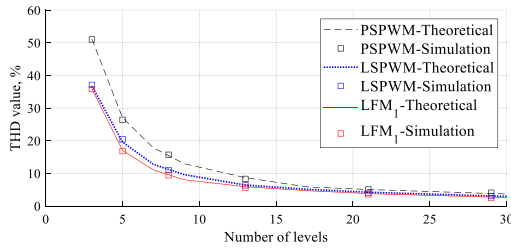
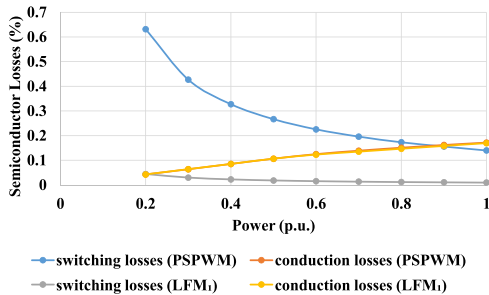
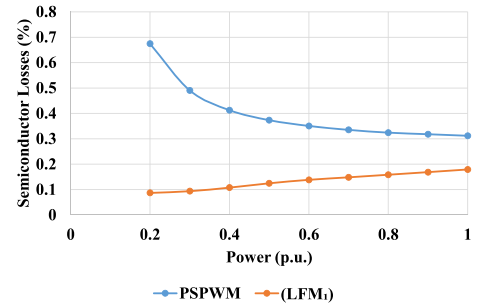
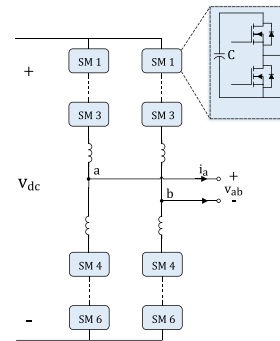

FIGURE 14. 13 – level simulation results under (a) PSPWM (b) LSPWM (c) LFM1.

TABLE 1. Simulation Parameters

Parameters	Description	Value
S	Rated Power	1 MVA
V_{llRMS}	Line-Line rms Voltage	13.8 kV
V_{dc}	dc Bus	24 kV
L_{grid}, R_{grid}	Grid Inductance, Resistance	3%, 1%
f_{fund}	Fundamental Frequency	60 Hz
f_c	PSPWM / LSPWM Carrier Frequency	5 kHz


FIGURE 15. Theoretical and simulation values for the line-to-line voltage THD versus the number of inverter voltage levels under low- and high-frequency modulation.

FIGURE 16. Switching and conduction losses under PSPWM and LFM1.

same as the carrier frequency, whereas the effective switching frequency of LSPWM is $f_c/2N$. The plots of Fig. 8 are at unity modulation index; the influence of carrier frequency and modulation index are later addressed in Section III-C2) and III-C3) respectively.


FIGURE 17. Total semiconductor losses under PSPWM and LFM1.

FIGURE 18. Circuit topology of the 4 – level single-phase MMC prototype.

It can be seen from the plot that for the same number of levels, the lowest THD is obtained with the modulation strategy with the least switching, that is, staircase modulation, or the low frequency modulation with one switching angle per quarter cycle, shown in Fig. 8 with the asterisk label (*), and referred to as LFM1. The harmonic distortion increases if the number of switching angles increases or if PSPWM or LSPWM are used. With more switching angles per quarter cycle, low-frequency modulation techniques LFM3, LFM5, and LFM7 accumulate high-frequency harmonics on top of the low-frequency harmonics, which make them worse from THD standpoint than LFM1 and carrier based modulation techniques. Also, the voltage THD of PSPWM is higher than that of LSPWM, owing to the higher effective switching

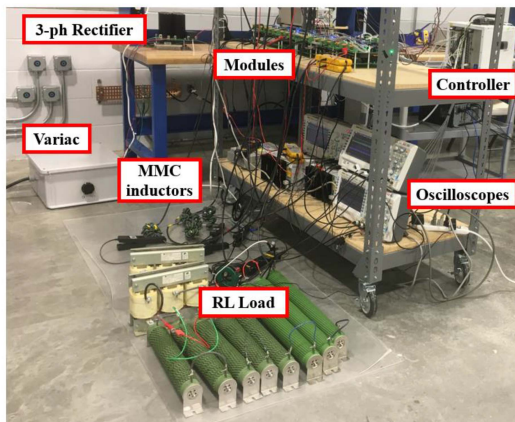


FIGURE 19. Experimental setup of the 4 – level MMC.

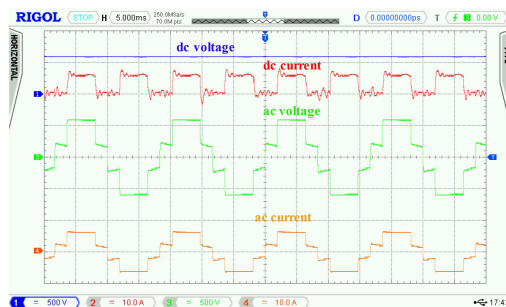


FIGURE 20. Experimental results under LFM₁.

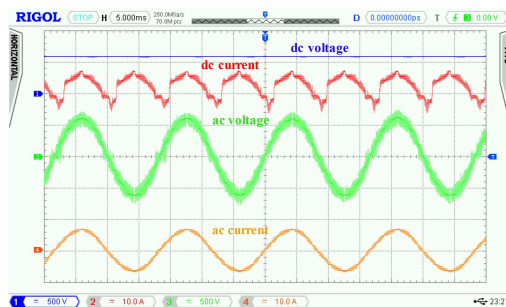


FIGURE 21. Experimental results under PSPWM.

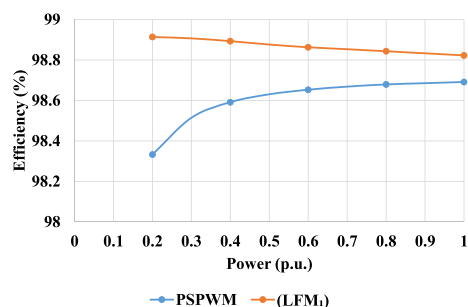


FIGURE 22. Efficiency under PSPWM and LFM₁.

frequency of the output PWM voltage of PSPWM for the same carrier frequency. These findings justify our claim that higher switching frequencies lead to higher voltage THD. It is important to clarify that those THD values obtained are for the line-to-line voltage at the output of the inverter. Hence, there is no effect of any harmonic filters whatsoever.

In grid applications, a THD < 5% is required by IEEE 519 standards [34]. According to Fig. 8, a MLI with at least 13-levels in its output phase voltage is required to meet the THD requirements, provided that the staircase modulation (LFM₁) is used. If PSPWM or LSPWM modulation methods are used, more voltage levels would be required to meet the same THD requirement, and capacitive and inductive passive harmonic filters would be needed to filter out the high frequency harmonics due to PWM switching. Although the harmonic filter size shrinks with higher switching frequency, a minimum harmonic filtering is still required nevertheless. On the contrary, it is possible to completely eliminate the need of bulky harmonic filters using staircase modulation with higher level inverters such as the CHB or MMC when these inverters are implemented with tens of levels.

Using staircase modulation in LV applications would not be possible since it would require high number of voltage levels to eliminate low frequency harmonics, which in turn means using more switching devices to synthesize more voltage levels. This fact justifies the need for high switching frequency converters with harmonic filters in such low and medium power inverter applications to get sinusoidal voltage and current waveforms with the smallest possible converter footprint. However, in MV applications, a high number of switching devices is required due to the device breakdown voltage limitation. With more switching modules, more voltage levels can be synthesized, and staircase modulation was shown to achieve the smallest THD at a given voltage level, which allows eliminating the need to voltage and current harmonic filters for higher level inverters. This fact might seem counterintuitive considering the tendency to use high switching frequency nowadays to achieve smaller magnetics.

Staircase modulation allows higher utilization of the DC bus voltage, and reduces the voltage stress on the devices. Using the emerging WBG devices with high breakdown voltages, such as the 1.7 kV, 3.3 kV, 6.5 kV, and 10 kV SiC modules allows a transformer-less connection of MV higher level inverters up to 69 kV distribution systems with only tens of switching devices, compared to hundreds needed with Si-IGBTs. Using tens of levels is also a rational approach since the improvement in THD is negligible beyond tens of voltage levels as shown in Fig. 8. Having a smaller number of voltage levels means reduced inverter switching devices count, reduced thermal management requirements, reduced cost, and improved efficiency and reliability.

2) EFFECT OF CARRIER FREQUENCY ON THD

It can be seen from (17) and (22) that the DFTS expression for PSPWM and LSPWM modulation are dependent on the

carrier frequency ω_c . The effect of the carrier frequency on THD of PSPWM and LSPWM is shown in Fig. 9. As shown in the figure, the voltage THD for PSPWM and LSPWM initially decreases with higher carrier frequency ratios $m_f = f_c/f_{fund}$, up to $m_f = 57$, beyond which THD variation becomes minimal. This fact can be explained by looking at the output voltage harmonic spectra of carrier-based modulated inverter shown in Fig. 10. For applications using very low carrier frequency ratios (say ≤ 13), carrier harmonics and their sidebands are significant compared to the fundamental component, and have a frequency below or equal to the fundamental frequency, which interferes with the fundamental component of the output voltage and deteriorates its quality. As f_c increases, carrier harmonics and their corresponding sideband harmonics are pushed away from the fundamental component, and spread over the higher frequency end of the spectrum. As a result, the rms equations given in (21) and (23) become less sensitive to change since B_m and C_{mn} coefficients become less significant, driven by the Bessel function properties. Since the THD is dependent directly on rms and fundamental component, a non-varying rms quantity results in a steady-state THD value.

Note that the discussion above on a stable voltage THD with higher carrier frequency ratios is only true for a filter-less inverter operation. When capacitive or inductive harmonic filters are used at the inverter output, high frequency carrier harmonics get attenuated by harmonic filters and less THD is achieved. Since passive LC harmonic filters shrink in size with higher switching frequency, using the same filter size with higher switching frequency results in more attenuation in harmonics and less THD. However, when no harmonic filter is used, there is no advantage in having higher switching, except having a slightly lower output current THD due to the filtering done by grid inductance in grid applications or the inductive nature of motor windings in drive applications.

3) EFFECT OF MODULATION INDEX ON THD

Fig. 11 shows the influence of modulation index (MI) on line-to-line voltage THD for different number of voltage levels. Higher THD is observed at lower MI. Note that LFM₁ is often implemented for inverters operating at high MI (>0.7). At lower MI, harmonic distortion increases significantly with staircase modulation since less voltage steps will be generated.

D. SIMULATION VERIFICATION

A simulation model has been built to verify the theoretical analysis on the THD. A 13-level 3-phase MMC whose topology shown in Fig. 12 is used; the dc side of the MMC is connected to a dc voltage source and the ac side is connected to the grid without any filter or transformer, as illustrated in Fig. 13. The simulation parameters are shown in Table 1. The simulation is performed under PSPWM, LSPWM, and staircase modulation.

1) RESULTS ANALYSIS

Fig. 14 shows the line-to-line voltage and the line current waveforms with 13 – level MMC under PSPWM, LSPWM, and staircase modulation. Under PSPWM, the line-to-line voltage THD is 8.32 %, much higher than the 5% requirement, although the current THD is low enough (0.78%). For LSPWM, the line-to-line voltage THD is lower at 6.4%, owing to less effective switching frequency for the same carrier frequency as compared to PSPWM. The current THD is at 1.26%. However, under staircase modulation, the theoretical analysis indicated that a THD $< 5\%$ can be achieved with 13 – level inverter. This is proven in simulation as the line-to-line voltage THD is greatly reduced to 4.76 %. Also, the line current THD is less than the 5% THD requirement (1.73%).

The THD results for the modulation strategies also agree with their theoretical values. The MMC is simulated under the same parameters shown in Table 1, but with a resistive load and with different output voltage levels. Fig. 15 shows a comparison between theoretical and simulation line-to-line voltage THD results for PSPWM, LSPWM, and staircase modulation with the 3, 5, 8, 13, 21, and 29 output phase voltage levels. The THD for simulated cases match the respective theoretical values, and validate the analytical study.

IV. LOW VERSUS HIGH PWM FREQUENCY – COMPARISON FROM THE EFFICIENCY PERSPECTIVE

Conduction and switching losses are the two types of losses associated with switching devices in a power electronics circuit. In this section, the impact of the PWM frequency on the power losses and efficiency is studied for the lowest and highest frequency modulation schemes, namely the staircase modulation and PSPWM at 10 kHz PWM frequency, respectively.

A. POWER LOSS COMPARISON

To evaluate the switching and conduction device power losses, a thermal model of the 1.2 kV SiC half-bridge module FF45MR12W1M1_B11 from Infineon is developed in PSIM software based on the turn on and turn off energies E_{on} and E_{off} , and the switching and conduction losses are evaluated and plotted in Fig. 16 for different power levels. As expected, the switching losses are much higher with PSPWM due to the higher switching frequency, and become more significant in terms of percentage at low power levels. At 70% of nominal power, switching losses are about 0.2% for PSPWM compared to 0.02% for staircase modulation. As for conduction loss, it is proportional to the device current and turn on resistance. Since the same rms current flows through the device under low and high-frequency modulation, the two schemes have the same conduction losses, around 0.18% at nominal power. Hence, the total losses are lower under staircase modulation (LFM₁) as shown in Fig. 17.

B. EXPERIMENTAL RESULTS AND EFFICIENCY COMPARISON

To visualize the influence of semiconductor losses on efficiency, a 1 kW, 600Vdc, 4-level, single – phase MMC prototype is built in the lab, and the converter is run under PSPWM and under staircase modulation. The MMC topology is shown in Fig. 18 and the prototype setup is shown in Fig. 19. The MMC is connected to an RL load with 1 p.u. resistance and 0.01 p.u. inductance. The results under staircase modulation and under PSPWM are shown in Figs. 20 and 21 respectively. The input and output powers are computed from the measured dc and ac voltages and currents, and the efficiency is plotted in Fig. 22.

It is evident that the staircase modulation achieves the highest efficiency, which peaks at 98.9%, compared to 98.7% for 10 kHz PSPWM. The experimental results show that the output voltage and current quality are better with PSPWM compared to staircase modulation, which at a first glance seems to contradict the theoretical analysis and the claim made earlier. However, the reason is that the prototype is implemented with 4-levels only, and the load inductor (2.5 mH) is acting as a passive low-pass filter for current harmonics with a cutoff frequency of ~ 6 kHz, which justifies why low current THD is observed with 10 kHz switching frequency PSPWM. Nevertheless, the advantage of staircase modulation in terms of THD appears with higher level inverters implemented with tens of voltage levels. Note that the effect of the ac – side inductor on the voltage THD is apparent when the inverter is implemented with few number of voltage levels as seen in Fig. 21, whereas it does not have the same significance when the inverter has high number of voltage levels, as seen in the simulation results.

V. CONCLUSION

The choice between high switching frequency and low switching frequency in power electronic converters has significant implications on the power quality, filter sizing, and efficiency. With the emergence of WBG devices, high-switching frequency has been prevailing nowadays, given the advantages it brings in terms of high power density power conversion in low and medium power applications, including dc-dc applications, SSTs, and low power inverters.

Notwithstanding the benefits of high-switching frequency modulation, this paper questions whether high frequency modulation achieves the least harmonic filter requirement for MV high power inverters implemented with high number of voltage levels. The paper analyzes the harmonic performance, filter requirements, and efficiency of low and high frequency modulation in MV, high power, higher level inverters. An analytical comparison of the multilevel inverter output voltage THD was performed under low and high frequency modulation. The analysis has shown that:

- 1) Higher switching frequency enables the use of smaller passive harmonic filters. Nevertheless, with no harmonic filters, higher switching frequency causes

more distortions in the output inverter voltage. Without the effect of harmonic filters, the lowest voltage THD is achievable with the modulation strategy with the least switching frequency, which is the lowest (fundamental) frequency modulation, also known as the staircase modulation.

- 2) A low THD ($< 5\%$) in both output voltage and current is achievable with the lowest frequency modulation with the least number of voltage levels. In other words, more voltage levels would be needed with high-frequency modulation to achieve the same harmonic performance as with low-frequency modulation.
- 3) Unlike LV inverters implemented with few levels, MV inverters are implemented to synthesize multi-level output voltage waveforms. With tens of levels, higher level inverters, specifically MMC and CHB inverters, can generate a series of staircase voltage levels that approximates a sinusoidal voltage without having to do high frequency switching. As a result, the need for both voltage and current filters can be eliminated and the lowest THD can be achieved with lowest frequency switching.

These conclusions might seem counterintuitive considering the tendency to use high switching frequency nowadays to achieve smaller magnetics.

Afterwards, the semiconductor power losses were compared for low – and high – frequency modulation. Although the conduction losses are same for the two modulation approaches, the switching, hence the total, losses are obviously much higher under high – frequency PWM.

Simulations have been carried out for a grid – connected MMC with different number of voltage levels, and the THD analysis was verified. Finally, experimental results on a 4 – level MMC were provided to compare converter efficiency under low- and high-frequency modulation approaches. The staircase modulation clearly displays the lower losses and higher efficiency, which makes it even more appealing for high power inverter applications.

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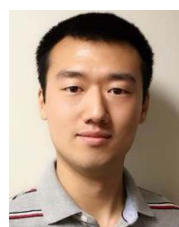
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